Reinforcement Learning Foundations

From Token Prediction to Outcome-Driven Learning

Deep Dive into Kimi K3 — Chapter 18

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Foundation Books Project

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Goal: differentiate expected reward, attach a valid comparison, keep each sample's behavior-policy identity, and calculate one complete update — with every setting illustrative, not a K3 trainer setting.

Roadmap

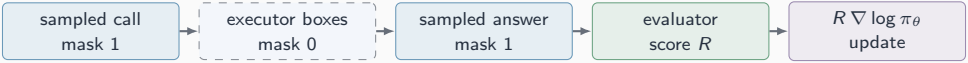
Outcomes train sampled probabilities

Comparisons and reuse

Reference algorithms and one complete update

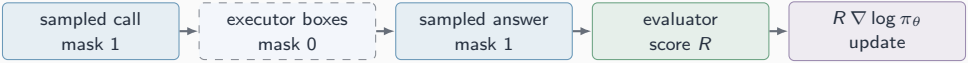
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- One-answer policy, $p = 1/4$: exact gradient $3/16$; one sampled blue at step 0.4 moves p to ≈ 0.3103. [DERIVED: illustrative fixture]

Comparisons and reuse

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- Behavior, current, and reference are separate roles; KL is directed (0.1308 versus 0.1438) and entropy is a different control.

Reference algorithms and one complete update

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Next: Chapter 19's Long-CoT Reinforcement Learning and Localized Policy Optimization.